

Does earnings information predict abnormal returns?

Ranking skill without a tradable edge, across 13,675 S&P 500 announcements, and the machinery built to tell those two apart.

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ABSTRACT

Post-earnings-announcement drift is among the most-studied anomalies in finance and among the easiest to reproduce spuriously. This note tests whether information in earnings releases predicts short-horizon abnormal returns across 13,675 announcements by 466 S&P 500 companies, timestamped to the minute from SEC filings, with each year from 2019 to 2024 held out and predicted by a model frozen before it began. **The answer splits.** Mean out-of-sample rank IC is 0.067 ($t = 3.40$) against -0.047 for standardised unexpected earnings alone, so the ranking is real. Nothing about it is tradable: the sector-neutral book returns a net Sharpe of -0.30 , alpha against Fama–French five factors plus momentum is -0.91% ($t = -1.11$), and running the same book on the same events 200 times with the predictions shuffled yields -0.70 on average, placing the realised value at the 84th percentile of noise ($p = 0.16$). One caveat qualifies the first half: the features are a median 83 days old at the announcement, so this is a claim about quarterly fundamentals rather than about earnings releases. The text of the releases, present for every event, carries 41% of the model's coefficient weight and moves out-of-sample skill by -0.0004 . Run on synthetic data with a known planted effect the same pipeline recovers it in 6/6 years at a net Sharpe of 4.93; with nothing planted it returns 0.024 and 0.48. That contrast is why the split above is worth believing rather than merely asserting.

1 · DESIGN

For each year Y , the model is fitted only on announcements whose 20-day outcome had fully resolved before 1 January Y , less a 25-session embargo; it is then frozen and used to predict every announcement in Y . The target is the cumulative abnormal return from **one session after** the announcement to twenty sessions after: windows beginning on the announcement day contain the opening gap, which cannot be traded, and booking it is the single largest source of overstated drift results. Events come from SEC 8-K Item 2.02 filings, so every one carries a minute-level acceptance timestamp rather than a vendor date — 100% of the sample has a declared, not assumed, before-open or after-close flag. Features are fundamental changes differenced year-on-year and standardised unexpected earnings, each stamped with the moment it became public; a single guard refuses any panel where a feature post-dates the trade. The text of every earnings press release -- exhibit 99.1 of the same 8-K -- is scored for tone, uncertainty and how much it has been rewritten since the firm's previous release, with the TF-IDF space fitted on an expanding window so no filing is weighted by vocabulary that appeared after it.

2 · SIX HELD-OUT YEARS

Year	Train n	Test n	IC	IC t	Pred. bp	Real. bp	Calib.	Sharpe
2019	4,876	1,348	0.039	1.50	180	26	0.10	-2.80
2020	6,223	1,399	0.123	5.67	155	268	1.56	1.26
2021	7,622	1,448	0.061	0.86	243	210	0.86	-0.43
2022	9,068	1,489	0.115	2.88	237	161	0.90	1.18
2023	10,559	1,534	-0.007	-0.31	250	-46	-0.13	-2.08
2024	12,089	1,562	0.072	1.34	205	67	0.24	-1.27

Table 1. Each row is a year the model never saw in training. *Pred.* and *Real.* are the predicted and realised top-minus-bottom quintile spreads in basis points; *Calib.* is the slope of realised on predicted, where 1.00 would mean the magnitudes were exactly right. The model predicted a positive spread of 155–250 bp every single year and realised between -46 and 268 : confident throughout, and over-confident in 5 of 6 years. Ranking and sizing are different skills, and the gap shows in 2019: a positive information coefficient and the worst realised profit of the six, because the calibration slope was 0.10.

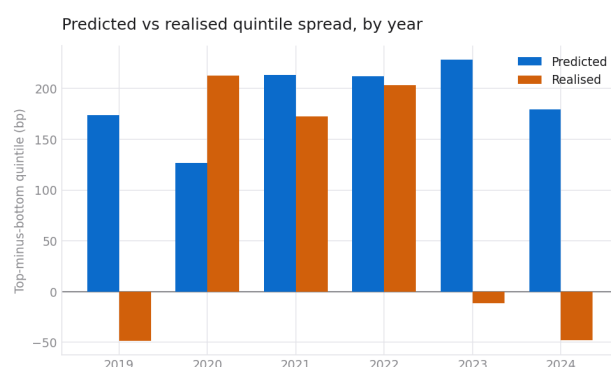


Figure 1. Predicted against realised top-minus-bottom quintile spread, by held-out year. A model with no signal still has opinions.

3 · WHY BELIEVE ANY OF IT

A real result and a broken pipeline look identical from outside, so the same code was run twice more on synthetic markets where the answer is known in advance. With a post-announcement drift planted in the data it is recovered in 6/6 years at a net Sharpe of 4.93. With nothing planted it returns 0.024 and 0.48 — yet still posts an IC of 0.080 at $t = 3.67$ in one individual year, a false positive in data where there is provably nothing to find. Anyone reporting a single year would have reported that one; the same caution applies to 2020 in Table 1.

Statistic	Real data	Synthetic null
Mean out-of-sample IC	0.067	0.024
t across the six years	3.40	1.24
Years with positive IC	5/6	4/6
Mean realised spread (bp)	114	103
Mean calibration slope	0.59	0.28
Net Sharpe, years stitched	-0.30	0.48
Newey–West t on daily P&L	-0.78	0.86

Table 2. The real study beside the synthetic null control. They look alike, which is the point: the real result is consistent with there being nothing to find.

4 · ALPHA, MULTIPLE TESTING, AND A SECOND METHOD

Regressed on Fama–French five factors plus momentum with Newey–West errors, the book shows an annualised alpha of -0.91% ($t = -1.11$) — no alpha. What the regression does find is significant loadings on profitability (RMW, $t = 3.62$), negative momentum (MOM, $t = -3.25$), value (HML, $t = 2.55$) and conservative investment (CMA, $t = 2.15$): the strategy is a quality-and-value portfolio in disguise, and those are compensated factors available for a few basis points. That alpha should also be read against the shuffled-prediction null in the abstract rather than against zero. Twelve specifications are logged, including the abandoned ones, giving a deflated Sharpe ratio of 0.00. Fama–MacBeth over 60 monthly cross-sections finds no feature significant at 5%, agreeing with the portfolio sort — which is the outcome that should raise confidence in a null rather than lower it.

5 · THE LIMITATION THAT MATTERS MOST

A point-in-time universe is not enough if the price source cannot serve delisted names. Yahoo returns no price history for 61% of the names deleted from the index during the sample, against 5% of the survivors, so 466 of the 735 members could be studied at all. The names lost include SIVB and FRC — Silicon Valley Bank and First Republic, both of which failed in 2023. Survivorship bias therefore re-enters through the data source even though the universe definition excludes it, and its direction is knowable: the missing firms are disproportionately those that collapsed, so the true result is probably somewhat worse than reported. Separately, SEC XBRL tags quarterly facts inconsistently. Differencing year-to-date flows into discrete quarters roughly doubled cash-flow coverage, but it still runs 40–50% rather than 100%. Deeper than either: the median feature is 83 days old at the announcement it is attached to, because the statements for the quarter being announced arrive with the 10-Q weeks later. Everything is point-in-time correct and almost nothing is about the release. Fixing the price source needs CRSP; fixing the staleness needs the figures parsed out of the release itself.

6 · REPRODUCIBILITY

make reproduce regenerates every number here and hashes each artefact; make verify diffs against the committed manifest and CI fails if a hash moves. Method, the falsification criteria fixed beforehand (two of five met), and a catalogue of the ways this study could flatter itself: github.com/pingadom/Earnings-Event-Research.